superior tool lands in the hands of the general public that can help them in unimaginable ways, such as creating a travel itinerary for a summer trip to Athens, summarizing a legal document in layman's terms, developing a self-help plan for losing weight, analyzing code for security and performance bugs, designing an application's data models, comparing and contrasting technologies for a specific use case, writing complaint letters to the Twitter CEO and much more.

ChatGPT (https://chat.openai.com) is a conversational agent (chatbot) built on a GPT (generative pretrained transformer) architecture, and it is capable of generating human-like text based on user prompts. It is an instance of a large language model (LLM) designed for conversation with a specific goal of generating safe, relevant content while engaging in a meaningful dialogue. The model is fed vast amounts of text data and learns to predict the next word in a sentence. It is trained using a diverse range of internet text, but it can also be fine-tuned with specific datasets for various tasks. Through this process, the model learns parts and parcels of human language text: grammar, punctuation, syntax, facts about the world, and some degree of reasoning ability.

> **NOTE** *LLM* is a broad term that refers to any large-scale model trained to understand or generate human-like text. These models are characterized by their vast number of parameters and their ability to handle a wide range of natural language processing tasks. LLMs can be based on various architectures and training methods.

With the release of ChatGPT to the public, a sudden race in the space of search using AI emerged overnight. ChatGPT has become a disruptor to many industries and is no less a threat to Google search. AI-backed tools similar to ChatGPT will disrupt many industries in the coming years. Under tremendous pressure—and possibly to save its search leader status—Google decided to unleash its version of conversational generative AI: its agent called Bard (https://bard.google.com) was made publicly available in May 2023.

In the meantime, Microsoft committed to investing $10 billion in ChatGPT, over and above its initial investment of $3 billion since 2019. Microsoft's Edge browser is integrated with ChatGPT via the Bing search engine, which also was made publicly available in May 2023. In addition, Microsoft rolled out AI-powered Microsoft 365 apps, so the AI agent is available in Microsoft Word, Excel, email, and other tools. Meta's LLaMA is another tool that has begun competing in the generative AI race.

The GPT-3 and -4 models were trained on billions of digital copies of books, articles, papers, blogs, and so on. The GPT-4 model was fed with data until September 2021 (it can't retrieve data after that date). Although there is no internet access for GPT-4 to fetch real-time information, as I write this, OpenAI just released a beta version of a web browser version for its Plus subscribers. So, I expect an internet-enabled generative AI assistant from OpenAI to be available to the general public soon.

Search engineers are being asked fundamental questions, including how generative AI agents will change the course of search. Let's answer this question by asking